In [1]:

```
!pip install spacy -q
import pandas as pd
import spacy
from collections import Counter

# Download and load the English spaCy model
!python -m spacy download en_core_web_sm -q

nlp = spacy.load("en_core_web_sm")

df = pd.read_csv("Reviews.csv")

print("Q1: Dataset Loaded Successfully")
display(df.head())


# Q2: Clean the dataset

df = df[['Text']].dropna()
df = df.drop_duplicates(subset=['Text'])
df = df.head(10000).copy()
df.reset_index(drop=True, inplace=True)

print("\nQ2: Dataset Cleaning")
print("Number of reviews:", len(df))
print("Missing values:", df['Text'].isnull().sum())
print("Duplicate reviews:", df['Text'].duplicated().sum())


# Q3: Apply spaCy and display tokens, lemmas, and POS tags for five reviews

print("\nQ3: Tokens, Lemmas and POS Tags")

for i in range(min(5, len(df))):
    doc = nlp(df.loc[i, 'Text'])

    print(f"\nReview {i + 1}")
    print("Original Review:", df.loc[i, 'Text'])
    print("Tokens:", [token.text for token in doc])
    print("Lemmas:", [token.lemma_ for token in doc])
    print("POS Tags:", [token.pos_ for token in doc])


# Q4: Identify nouns, verbs, adjectives, and adverbs

print("\nQ4: POS Categories")

doc = nlp(df.loc[0, 'Text'])

nouns = [token.text for token in doc if token.pos_ == 'NOUN']
verbs = [token.text for token in doc if token.pos_ == 'VERB']
adjectives = [token.text for token in doc if token.pos_ == 'ADJ']
adverbs = [token.text for token in doc if token.pos_ == 'ADV']

print("Nouns:", nouns)
print("Verbs:", verbs)
print("Adjectives:", adjectives)
print("Adverbs:", adverbs)
```

```python
# Q5: Count the frequency of major POS categories in all reviews

pos_counts = Counter()

for doc in nlp.pipe(df['Text'].astype(str), batch_size=100):
    for token in doc:
        if token.pos_ in ['NOUN', 'VERB', 'ADJ', 'ADV']:
            pos_counts[token.pos_] += 1

pos_df = pd.DataFrame(
    pos_counts.items(),
    columns=['POS Category', 'Frequency']
)

print("\nQ5: POS Category Frequencies")
display(pos_df)


# Q6: Named Entity Recognition (NER)

entity_records = []

for i, doc in enumerate(
    nlp.pipe(df['Text'].astype(str), batch_size=100)
):
    for ent in doc.ents:
        if ent.label_ in [
            'PERSON', 'ORG', 'DATE', 'MONEY',
            'QUANTITY', 'PRODUCT'
        ]:
            entity_records.append({
                'Review ID': i + 1,
                'Review': df.iloc[i]['Text'],
                'Entity': ent.text,
                'Entity Label': ent.label_
            })

entity_df = pd.DataFrame(
    entity_records,
    columns=['Review ID', 'Review', 'Entity', 'Entity Label']
)

print("\nQ6: Extracted Named Entities")
display(entity_df.head(30))

print("\nTotal entities extracted:", len(entity_df))


# Q7: Create an information extraction function

def extract_information(review):
    doc = nlp(review)

    tokens = [token.text for token in doc]
    lemmas = [token.lemma_ for token in doc]
    pos_tags = [token.pos_ for token in doc]

    named_entities = [
```

```
        {
            'Entity': ent.text,
            'Entity Label': ent.label_
        }
        for ent in doc.ents
    ]

    return {
        'Tokens': tokens,
        'Lemmas': lemmas,
        'POS Tags': pos_tags,
        'Named Entities': named_entities
    }


# Q8: Test the function using three new customer reviews

test_reviews = [
    "I bought two packets of organic coffee from Starbucks for $15 yesterday.",
    "John purchased a new chocolate product from Amazon in January 2025.",
    "The company Nestle delivered five boxes of cookies to London last week."
]

print("\nQ8: Testing Information Extraction Function")

for i, review in enumerate(test_reviews, start=1):
    print(f"\nTest Review {i}: {review}")
    result = extract_information(review)

    print("Tokens:", result['Tokens'])
    print("Lemmas:", result['Lemmas'])
    print("POS Tags:", result['POS Tags'])
    print("Named Entities:", result['Named Entities'])


# Q9: Display extracted information in a structured DataFrame

structured_records = []

for i, review in enumerate(test_reviews, start=1):
    result = extract_information(review)

    structured_records.append({
        'Review ID': i,
        'Review': review,
        'Tokens': ', '.join(result['Tokens']),
        'Lemmas': ', '.join(result['Lemmas']),
        'POS Tags': ', '.join(result['POS Tags']),
        'Named Entities': ', '.join(
            ent['Entity'] for ent in result['Named Entities']
        ),
        'Entity Labels': ', '.join(
            ent['Entity Label'] for ent in result['Named Entities']
        )
    })

structured_df = pd.DataFrame(structured_records)

print("\nQ9: Structured Information Extraction Results")
display(structured_df)
```

```
# FINAL CONCLUSION

print("\nCONCLUSION")
print("spaCy was used to process customer reviews and extract tokens,")
print("lemmas, POS tags, and named entities.")
print("The extracted information was organized into DataFrames")
print("for further customer feedback analysis.")
```

```
[+] Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
Q1: Dataset Loaded Successfully
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	Helpfulness
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1	
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa	0	
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"	1	
3	4	B000UA0QIQ	A395BORC6FGVXV	Karl	3	
4	5	B006K2ZZ7K	A1UQRSCLF8GW1T	Michael D. Bigham "M. Wassir"	0	



Q2: Dataset Cleaning
Number of reviews: 10000
Missing values: 0
Duplicate reviews: 0

Q3: Tokens, Lemmas and POS Tags

Review 1
Original Review: I have bought several of the Vitality canned dog food products a
nd have found them all to be of good quality. The product looks more like a stew
than a processed meat and it smells better. My Labrador is finicky and she apprec
iates this product better than most.
Tokens: ['I', 'have', 'bought', 'several', 'of', 'the', 'Vitality', 'canned', 'do
g', 'food', 'products', 'and', 'have', 'found', 'them', 'all', 'to', 'be', 'of',
'good', 'quality', '.', 'The', 'product', 'looks', 'more', 'like', 'a', 'stew',
'than', 'a', 'processed', 'meat', 'and', 'it', 'smells', 'better', '.', 'My', 'La
brador', 'is', 'finicky', 'and', 'she', 'appreciates', 'this', 'product', 'bette
r', 'than', ' ', 'most', '.']
Lemmas: ['I', 'have', 'buy', 'several', 'of', 'the', 'Vitality', 'can', 'dog', 'f
ood', 'product', 'and', 'have', 'find', 'they', 'all', 'to', 'be', 'of', 'good',
'quality', '.', 'the', 'product', 'look', 'more', 'like', 'a', 'stew', 'than',
'a', 'process', 'meat', 'and', 'it', 'smell', 'well', '.', 'my', 'Labrador', 'b
e', 'finicky', 'and', 'she', 'appreciate', 'this', 'product', 'well', 'than', '
', 'most', '.']
POS Tags: ['PRON', 'AUX', 'VERB', 'ADJ', 'ADP', 'DET', 'PROPN', 'VERB', 'NOUN',
'NOUN', 'NOUN', 'CCONJ', 'AUX', 'VERB', 'PRON', 'PRON', 'PART', 'AUX', 'ADP', 'AD
J', 'NOUN', 'PUNCT', 'DET', 'NOUN', 'VERB', 'ADV', 'ADP', 'DET', 'NOUN', 'ADP',
'DET', 'VERB', 'NOUN', 'CCONJ', 'PRON', 'VERB', 'ADV', 'PUNCT', 'PRON', 'PROPN',
'AUX', 'ADJ', 'CCONJ', 'PRON', 'VERB', 'DET', 'NOUN', 'ADV', 'ADP', 'SPACE', 'AD
J', 'PUNCT']

Review 2
Original Review: Product arrived labeled as Jumbo Salted Peanuts...the peanuts we
re actually small sized unsalted. Not sure if this was an error or if the vendor
intended to represent the product as "Jumbo".
Tokens: ['Product', 'arrived', 'labeled', 'as', 'Jumbo', 'Salted', 'Peanuts',
'...', 'the', 'peanuts', 'were', 'actually', 'small', 'sized', 'unsalted', '.',
'Not', 'sure', 'if', 'this', 'was', 'an', 'error', 'or', 'if', 'the', 'vendor',
'intended', 'to', 'represent', 'the', 'product', 'as', '"', 'Jumbo', '"', '.']
Lemmas: ['product', 'arrived', 'label', 'as', 'Jumbo', 'Salted', 'Peanuts',
'...', 'the', 'peanut', 'be', 'actually', 'small', 'sized', 'unsalte', '.', 'no
t', 'sure', 'if', 'this', 'be', 'an', 'error', 'or', 'if', 'the', 'vendor', 'inte
nd', 'to', 'represent', 'the', 'product', 'as', '"', 'Jumbo', '"', '.']
POS Tags: ['NOUN', 'AUX', 'VERB', 'ADP', 'PROPN', 'PROPN', 'PROPN', 'PUNCT', 'DE
T', 'NOUN', 'AUX', 'ADV', 'ADJ', 'ADJ', 'VERB', 'PUNCT', 'PART', 'ADJ', 'SCONJ',
'PRON', 'AUX', 'DET', 'NOUN', 'CCONJ', 'SCONJ', 'DET', 'NOUN', 'VERB', 'PART', 'V
ERB', 'DET', 'NOUN', 'ADP', 'PUNCT', 'PROPN', 'PUNCT', 'PUNCT']

Review 3
Original Review: This is a confection that has been around a few centuries. It i
s a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cu
t into tiny squares and then liberally coated with powdered sugar. And it is a t
iny mouthful of heaven. Not too chewy, and very flavorful. I highly recommend t
his yummy treat. If you are familiar with the story of C.S. Lewis' "The Lion, Th
e Witch, and The Wardrobe" - this is the treat that seduces Edmund into selling o
ut his Brother and Sisters to the Witch.
Tokens: ['This', 'is', 'a', 'confection', 'that', 'has', 'been', 'around', 'a',
'few', 'centuries', '.', ' ', 'It', 'is', 'a', 'light', ',', 'pillowy', 'citrus',
'gelatin', 'with', 'nuts', '-', 'in', 'this', 'case', 'Filberts', '.', 'And', 'i
t', 'is', 'cut', 'into', 'tiny', 'squares', 'and', 'then', 'liberally', 'coated',

```
'with', 'powdered', 'sugar', '.', ' ', 'And', 'it', 'is', 'a', 'tiny', 'mouthfu
l', 'of', 'heaven', '.', ' ', 'Not', 'too', 'chewy', ',', 'and', 'very', 'flavorf
ul', '.', ' ', 'I', 'highly', 'recommend', 'this', 'yummy', 'treat', '.', ' ', 'I
f', 'you', 'are', 'familiar', 'with', 'the', 'story', 'of', 'C.S.', 'Lewis', "'",
'"', 'The', 'Lion', ',', 'The', 'Witch', ',', 'and', 'The', 'Wardrobe', '"', '-',
'this', 'is', 'the', 'treat', 'that', 'seduces', 'Edmund', 'into', 'selling', 'ou
t', 'his', 'Brother', 'and', 'Sisters', 'to', 'the', 'Witch', '.']
Lemmas: ['this', 'be', 'a', 'confection', 'that', 'have', 'be', 'around', 'a', 'f
ew', 'century', '.', ' ', 'it', 'be', 'a', 'light', ',', 'pillowy', 'citrus', 'ge
latin', 'with', 'nut', '-', 'in', 'this', 'case', 'filbert', '.', 'and', 'it', 'b
e', 'cut', 'into', 'tiny', 'square', 'and', 'then', 'liberally', 'coat', 'with',
'powdered', 'sugar', '.', ' ', 'and', 'it', 'be', 'a', 'tiny', 'mouthful', 'of',
'heaven', '.', ' ', 'not', 'too', 'chewy', ',', 'and', 'very', 'flavorful', '.',
' ', 'I', 'highly', 'recommend', 'this', 'yummy', 'treat', '.', ' ', 'if', 'you',
'be', 'familiar', 'with', 'the', 'story', 'of', 'C.S.', 'Lewis', "'", '"', 'the',
'Lion', ',', 'the', 'Witch', ',', 'and', 'the', 'Wardrobe', '"', '-', 'this', 'b
e', 'the', 'treat', 'that', 'seduce', 'Edmund', 'into', 'sell', 'out', 'his', 'br
other', 'and', 'Sisters', 'to', 'the', 'Witch', '.']
POS Tags: ['PRON', 'AUX', 'DET', 'NOUN', 'PRON', 'AUX', 'AUX', 'ADP', 'DET', 'AD
J', 'NOUN', 'PUNCT', 'SPACE', 'PRON', 'AUX', 'DET', 'ADJ', 'PUNCT', 'ADJ', 'NOU
N', 'VERB', 'ADP', 'NOUN', 'PUNCT', 'ADP', 'DET', 'NOUN', 'NOUN', 'PUNCT', 'CCON
J', 'PRON', 'AUX', 'VERB', 'ADP', 'ADJ', 'NOUN', 'CCONJ', 'ADV', 'ADV', 'VERB',
'ADP', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'CCONJ', 'PRON', 'AUX', 'DET', 'ADJ', 'AD
J', 'ADP', 'PROPN', 'PUNCT', 'SPACE', 'PART', 'ADV', 'NOUN', 'PUNCT', 'CCONJ', 'A
DV', 'ADJ', 'PUNCT', 'SPACE', 'PRON', 'ADV', 'VERB', 'DET', 'ADJ', 'NOUN', 'PUNC
T', 'SPACE', 'SCONJ', 'PRON', 'AUX', 'ADJ', 'ADP', 'DET', 'NOUN', 'ADP', 'PROPN',
'PROPN', 'PUNCT', 'PUNCT', 'DET', 'PROPN', 'PUNCT', 'DET', 'PROPN', 'PUNCT', 'CCO
NJ', 'DET', 'PROPN', 'PUNCT', 'PUNCT', 'PRON', 'AUX', 'DET', 'NOUN', 'PRON', 'VER
B', 'PROPN', 'ADP', 'VERB', 'ADP', 'PRON', 'NOUN', 'CCONJ', 'PROPN', 'ADP', 'DE
T', 'PROPN', 'PUNCT']

Review 4
Original Review: If you are looking for the secret ingredient in Robitussin I bel
ieve I have found it.  I got this in addition to the Root Beer Extract I ordered
(which was good) and made some cherry soda.  The flavor is very medicinal.
Tokens: ['If', 'you', 'are', 'looking', 'for', 'the', 'secret', 'ingredient', 'i
n', 'Robitussin', 'I', 'believe', 'I', 'have', 'found', 'it', '.', ' ', 'I', 'go
t', 'this', 'in', 'addition', 'to', 'the', 'Root', 'Beer', 'Extract', 'I', 'order
ed', '(', 'which', 'was', 'good', ')', 'and', 'made', 'some', 'cherry', 'soda',
'.', ' ', 'The', 'flavor', 'is', 'very', 'medicinal', '.']
Lemmas: ['if', 'you', 'be', 'look', 'for', 'the', 'secret', 'ingredient', 'in',
'Robitussin', 'I', 'believe', 'I', 'have', 'find', 'it', '.', ' ', 'I', 'get', 't
his', 'in', 'addition', 'to', 'the', 'Root', 'Beer', 'Extract', 'I', 'order',
'(', 'which', 'be', 'good', ')', 'and', 'make', 'some', 'cherry', 'soda', '.', '
', 'the', 'flavor', 'be', 'very', 'medicinal', '.']
POS Tags: ['SCONJ', 'PRON', 'AUX', 'VERB', 'ADP', 'DET', 'ADJ', 'NOUN', 'ADP', 'P
ROPN', 'PRON', 'VERB', 'PRON', 'AUX', 'VERB', 'PRON', 'PUNCT', 'SPACE', 'PRON',
'VERB', 'PRON', 'ADP', 'NOUN', 'ADP', 'DET', 'PROPN', 'PROPN', 'PROPN', 'PRON',
'VERB', 'PUNCT', 'PRON', 'AUX', 'ADJ', 'PUNCT', 'CCONJ', 'VERB', 'DET', 'NOUN',
'NOUN', 'PUNCT', 'SPACE', 'DET', 'NOUN', 'AUX', 'ADV', 'ADJ', 'PUNCT']

Review 5
Original Review: Great taffy at a great price.  There was a wide assortment of yu
mmy taffy.  Delivery was very quick.  If your a taffy lover, this is a deal.
Tokens: ['Great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'There', 'was',
'a', 'wide', 'assortment', 'of', 'yummy', 'taffy', '.', ' ', 'Delivery', 'was',
'very', 'quick', '.', ' ', 'If', 'your', 'a', 'taffy', 'lover', ',', 'this', 'i
s', 'a', 'deal', '.']
Lemmas: ['great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'there', 'be',
'a', 'wide', 'assortment', 'of', 'yummy', 'taffy', '.', ' ', 'Delivery', 'be', 'v
```

```
ery', 'quick', '.', ' ', 'if', 'your', 'a', 'taffy', 'lover', ',', 'this', 'be',
'a', 'deal', '.']
POS Tags: ['ADJ', 'NOUN', 'ADP', 'DET', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PRON',
'VERB', 'DET', 'ADJ', 'NOUN', 'ADP', 'ADJ', 'NOUN', 'PUNCT', 'SPACE', 'PROPN', 'A
UX', 'ADV', 'ADJ', 'PUNCT', 'SPACE', 'SCONJ', 'PRON', 'DET', 'PROPN', 'NOUN', 'PU
NCT', 'PRON', 'AUX', 'DET', 'NOUN', 'PUNCT']

Q4: POS Categories
Nouns: ['dog', 'food', 'products', 'quality', 'product', 'stew', 'meat', 'produc
t']
Verbs: ['bought', 'canned', 'found', 'looks', 'processed', 'smells', 'appreciate
s']
Adjectives: ['several', 'good', 'finicky', 'most']
Adverbs: ['more', 'better', 'better']

Q5: POS Category Frequencies
```

	POS Category	Frequency
0	VERB	95797
1	ADJ	71482
2	NOUN	144487
3	ADV	50615

```
Q6: Extracted Named Entities
```

	Review ID	Review	Entity	Entity Label
0	1	I have bought several of the Vitality canned d...	Vitality	ORG
1	1	I have bought several of the Vitality canned d...	My Labrador	PERSON
2	2	Product arrived labeled as Jumbo Salted Peanut...	Jumbo Salted	ORG
3	3	This is a confection that has been around a fe...	around a few centuries	DATE
4	3	This is a confection that has been around a fe...	pillowy citrus gelatin	ORG
5	3	This is a confection that has been around a fe...	C.S. Lewis'	ORG
6	3	This is a confection that has been around a fe...	Edmund	PRODUCT
7	3	This is a confection that has been around a fe...	Sisters	PERSON
8	3	This is a confection that has been around a fe...	Witch	ORG
9	5	Great taffy at a great price. There was a wid...	Delivery	PERSON
10	6	I got a wild hair for taffy and ordered this f...	five pound	QUANTITY
11	6	I got a wild hair for taffy and ordered this f...	only two weeks	DATE
12	7	This saltwater taffy had great flavors and was...	Fralinger's	ORG
13	9	Right now I'm mostly just sprouting this so my...	Wheatgrass	PERSON
14	11	I don't know if it's the cactus or the tequila...	Tequila Picante Gourmet de Inclan	ORG
15	13	My cats have been happily eating Felidae Plati...	Felidae Platinum	ORG
16	13	My cats have been happily eating Felidae Plati...	more than two years	DATE
17	14	good flavor! these came securely packed... the...	Twizzlers	ORG
18	15	The Strawberry Twizzlers are my guilty pleasur...	The Strawberry Twizzlers	ORG
19	15	The Strawberry Twizzlers are my guilty pleasur...	Six pounds	QUANTITY

	Review ID	Review	Entity	Entity Label
20	15	The Strawberry Twizzlers are my guilty pleasur...	I.	ORG
21	16	My daughter loves twizzlers and this shipment ...	six pounds	MONEY
22	18	I am very satisfied with my Twizzler purchase....	Twizzler	ORG
23	19	Twizzlers, Strawberry my childhood favorite ca...	Strawberry	PERSON
24	19	Twizzlers, Strawberry my childhood favorite ca...	Lancaster Pennsylvania	ORG
25	19	Twizzlers, Strawberry my childhood favorite ca...	Y & S Candies, Inc.	ORG
26	19	Twizzlers, Strawberry my childhood favorite ca...	the Hershey Company	ORG
27	19	Twizzlers, Strawberry my childhood favorite ca...	Company	ORG
28	19	Twizzlers, Strawberry my childhood favorite ca...	1845	DATE
29	19	Twizzlers, Strawberry my childhood favorite ca...	Young	ORG

```
Total entities extracted: 16012

Q8: Testing Information Extraction Function

Test Review 1: I bought two packets of organic coffee from Starbucks for $15 yest
erday.
Tokens: ['I', 'bought', 'two', 'packets', 'of', 'organic', 'coffee', 'from', 'Sta
rbucks', 'for', '$', '15', 'yesterday', '.']
Lemmas: ['I', 'buy', 'two', 'packet', 'of', 'organic', 'coffee', 'from', 'Starbuc
ks', 'for', '$', '15', 'yesterday', '.']
POS Tags: ['PRON', 'VERB', 'NUM', 'NOUN', 'ADP', 'ADJ', 'NOUN', 'ADP', 'PROPN',
'ADP', 'SYM', 'NUM', 'NOUN', 'PUNCT']
Named Entities: [{'Entity': 'two', 'Entity Label': 'CARDINAL'}, {'Entity': 'Starb
ucks', 'Entity Label': 'ORG'}, {'Entity': '15', 'Entity Label': 'MONEY'}, {'Entit
y': 'yesterday', 'Entity Label': 'DATE'}]

Test Review 2: John purchased a new chocolate product from Amazon in January 202
5.
Tokens: ['John', 'purchased', 'a', 'new', 'chocolate', 'product', 'from', 'Amazo
n', 'in', 'January', '2025', '.']
Lemmas: ['John', 'purchase', 'a', 'new', 'chocolate', 'product', 'from', 'Amazo
n', 'in', 'January', '2025', '.']
POS Tags: ['PROPN', 'VERB', 'DET', 'ADJ', 'NOUN', 'NOUN', 'ADP', 'PROPN', 'ADP',
'PROPN', 'NUM', 'PUNCT']
Named Entities: [{'Entity': 'John', 'Entity Label': 'PERSON'}, {'Entity': 'Amazo
n', 'Entity Label': 'ORG'}, {'Entity': 'January 2025', 'Entity Label': 'DATE'}]

Test Review 3: The company Nestle delivered five boxes of cookies to London last
week.
Tokens: ['The', 'company', 'Nestle', 'delivered', 'five', 'boxes', 'of', 'cookie
s', 'to', 'London', 'last', 'week', '.']
Lemmas: ['the', 'company', 'Nestle', 'deliver', 'five', 'box', 'of', 'cooky', 't
o', 'London', 'last', 'week', '.']
POS Tags: ['DET', 'NOUN', 'PROPN', 'VERB', 'NUM', 'NOUN', 'ADP', 'NOUN', 'ADP',
'PROPN', 'ADJ', 'NOUN', 'PUNCT']
Named Entities: [{'Entity': 'Nestle', 'Entity Label': 'ORG'}, {'Entity': 'five',
'Entity Label': 'CARDINAL'}, {'Entity': 'London', 'Entity Label': 'GPE'}, {'Entit
y': 'last week', 'Entity Label': 'DATE'}]

Q9: Structured Information Extraction Results
```

	Review ID	Review	Tokens	Lemmas	POS Tags	Named Entities	Entity Labels
0	1	I bought two packets of organic coffee from St...	I, bought, two, packets, of, organic, coffee, ...	I, buy, two, packet, of, organic, coffee, from...	PRON, VERB, NUM, NOUN, ADP, ADJ, NOUN, ADP, PR...	two, Starbucks, 15, yesterday	CARDINAL, ORG, MONEY, DATE
1	2	John purchased a new chocolate product from Am...	John, purchased, a, new, chocolate, product, f...	John, purchase, a, new, chocolate, product, fr...	PROPN, VERB, DET, ADJ, NOUN, NOUN, ADP, PROPN,...	John, Amazon, January 2025	PERSON, ORG, DATE
2	3	The company Nestle delivered five boxes of coo...	The, company, Nestle, delivered, five, boxes, ...	the, company, Nestle, deliver, five, box, of, ...	DET, NOUN, PROPN, VERB, NUM, NOUN, ADP, NOUN, ...	Nestle, five, London, last week	ORG, CARDINAL, GPE, DATE

```
CONCLUSION
spaCy was used to process customer reviews and extract tokens,
lemmas, POS tags, and named entities.
The extracted information was organized into DataFrames
for further customer feedback analysis.
```

In [ ]: